

Parsa Jafarian

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EDUCATION

McGill University

August 2024 – April 2028

B.Eng in Software Engineering Co-op – Math Minor | GPA: 3.94/4.00

Coursework: (Hons.) Stochastic Processes, Mathematical Finance, Applied ML, Theory of ML, Analysis I

EXPERIENCE

Machine Learning Research Intern

January – August 2026

McGill Networks Research Lab – Supervisor: Mark Coates

- Improving **SLiDE**, a Koopman-based RNN extending *SKOLR* with exogenous inputs, for cross-sectional equity return prediction, following the setup of *Deep Learning Statistical Arbitrage*
- Trained **SLiDE** on a **Sharpe Ratio** objective, achieving the best SR (**1.35**) among all baselines on $K=1$ PCA residuals, with a Kendall rank correlation of **0.003** confirmed significant via stationary bootstrap ($p < 0.001$)
- Built **PyTorch** data loader handling rolling window training, S&P500 memberships and covariates; validated against the original DLSA codebase, matching CNN-Transformer performance (SR **1.68** vs. their **1.51**)
- Explored a multi-scale **SLiDE** variant attending over multiple lookback horizons (daily, weekly, monthly), finding it underperformed the single-scale model (SR: **0.51** vs. **1.35**)

Robotics Research Assistant

September – December 2025

McGill NeuRoC Lab – Supervisor: Guillaume Durandau

- First author** extended abstract accepted at the International Conference on Neuro-Rehabilitation (ICNR) 2026
- Developed a **MuJoCo** simulation framework and a **SciPy**-based admittance-impedance controller for tremor mitigation in human-robot object co-manipulation
- Evaluated controller using RMSE and frequency magnitude, achieving **98% attenuation** and **4–14% RMSE**

Data Engineering Intern

January – August 2025

Transport Canada

- Migrated 9 **GeoPandas** pipelines to **PySpark** on **Databricks**, cutting annual costs by **\$24K**
- Optimized a **Power BI** dataset via Dataflows and DirectQuery, cutting load time from **2 hours to 20 minutes**

CLUBS & PROJECTS

Algorithmic Trading Member, McGill Quantitative Research: Investigated triangular arbitrage across 19 FX pairs using millisecond tick data; applied cointegration tests and OLS to identify mean-reverting spreads.

Commodity Prediction Algorithm, CodeML: Won 1st place building a commodity demand model on Statistics Canada data; Random Forest achieved 93.48% R^2 , 19.06% SMAPE.

AWARDS

Research Funding: NSERC Discovery Grant (*NSERC*) | Dan Adamson CDSI Support Fund (*McGill CDSI*)

Scholarships/Honours: Faculty of Engineering Scholarship | Dean's Honour List (*McGill*)

Hackathons: 3x winner, recognized at CodeML, JACHacks, and MariHacks

SKILLS

Languages: Python, SQL, Bash, C, Java, OCaml

Data Science: PyTorch, NumPy, Pandas, Scikit-Learn, SciPy, Matplotlib

Systems/Tools: Linux, Docker, Azure, Databricks, Git, Latex